MATHEMATICS CLASS – X WBBSE 2024

1. Choose the correct option in each case from the following: (1X6=6)

   i. If the rate of simple interest and compound interest are both 10% per annum then the ratio of simple interest and compound interest in second year of a certain principal will be:
   (a) 20:21 (b) 10:11 (c) 5:6 (d) 1:1

   ii. If one root of the equation $ax^2 + abcx + bc = 0, a \neq 0$ is reciprocal to the other, then
   (a) abc=1 (b) b = ac (c) bc = a (d) a = bc

   iii. Two circles of diameter 5 cm and 7 cm touch each other internally, the distance between their centres will be:
   (a) 1 cm (b) 2 cm (c) 3 cm (d) 4 cm

   iv. The maximum value of $tan\theta + cot\theta$ is :
   (a) 0 (b) 2 (c) -2 (d) 1

   v. If the height and base area of a solid hemisphere and a solid cylinder are same, then the ratio of their volume is:
   (a) 1:3 (b) 1:2 (c) 2:3 (d) 3:4

   vi. If A and M be the arithmetic mean and median of first ten natural numbers, then relation between them will be:
   (a) A>M (b) A<M (c) $A = \frac{1}{M}$ (d) A = M

2. Fill in the blanks (any five):- (1X5=5)

   i. The equation $(P-3)x^2 + 5x + 10 = 0$ will not be a quadratic equation for P =___ |
   ii. The sum of Principal and Compound interest in certain time is called ______ |
   iii. The corresponding sides of two similar triangles are __________ |
   iv. If $sin(\theta - 30^0) = \frac{1}{2}$ then $cos\theta =$________ |
   v. If V be the volume, R be the radius of the base and H be the height of a right circular cone , then H =_______ |
   vi. The numbers 8, 9, 12, 17, x+2, x+4, 30, 34, 39 are in increasing order and if the median of these numbers is 24, then x = ____________ ।

3. Write True or False (any five):- (1X5=5)

   i. In a partnership business, the ratio of the capitals of three members is a:b:c, and the ratio of time of investment of their capitals is x:y:z. The ratio of their profit will be ax:by:cz,
   ii. If $a \propto b, b \propto \frac{1}{c}$ and $c \propto d$ then $a \propto \frac{1}{d}$.
   iii. If two chords of a circle are at an equal distance from the centre of that circle, then those two chords will be always equal.
   iv. The hour hand of a clock rotates through an angle $\frac{\pi}{6}$ radians in 2 hours.
   v. The ratio of whole surfaces of a solid sphere and a solid hemisphere of equal radius is 2:1.
   vi. The average of n numbers is $\overline{x}$. If the sum of first (n-1) numbers is K, then nth number is $(n-1)\overline{x} + K$

4. Answer the following questions (any ten):- (2X10=20)

i. In how many years will Rs. 500 get Rs. 105 compound interest at 10% per annum compounded annually.

ii. In a partnership business, ratio of capitals of Ila, Rahima and Bela is 3:8:5. If profit of Ila is Rs, 600 less than the profit of Bela, then find the total profit in the business.

iii. If the roots of the equation $x^2 - 22x + 105 = 0$ are α and β then find the value of $\frac{1}{\alpha}+\frac{1}{\beta}$.

iv. If (3x-2y):(3x+2y) = 4:5 , then find the value of (x+y):(x-y) .

v. BOC is a diameter of a circle whose centre is O. ABCD is a cyclic quadrilateral and if $\angle ADC = 110^0$ then find $\angle ACB$.

vi. In a trapezium ABCD , BC||AD and AD = 4 cm, two diagonals AC and BD meet at O in such a way that $\frac{AO}{OC}=\frac{DO}{OB}=\frac{1}{2}$, then find the length of BC.

vii. In $\triangle$ABC , $\angle ABC = 90^0$, AB = 6 cm, BC = 8 cm then find the circumradius of $\triangle ABC$.

viii. If $r\cos\theta = 2\sqrt{3}, \quad r\sin\theta = 2$, then find the value of r and θ when $\left(0^0 < \theta < 90^0\right)$.

ix. If sin(A+B)=1, and cos(A-B)=1, then find the value of cot 2A, $0^0 \le A + B \le 90^0$, $A \ge B$.

x. How much percent of the curved surface area of a sphere will be increased if the radius is doubled?

xi. The length of the diagonal of each face of a cube is $6\sqrt{2}$ cm. Find the total surface area of the cube.

xii. The mean of a frequency distribution is 7 and $\sum f_i x_i$ = 140, then find the value of $\sum f_i$.

5. Answer any one question:- (1X5=5)

i. After retirement of his service, Gobinda Babu got Rs. 500000. He deposited a part of it in post office at 7.2 % simple interest per annum and he deposited other part in a bank at 6% simple interest per annum. Every year he got Rs 33600 in total as interest from bank and post office. Find the amounts he deposited in bank and post office.

ii. Aman has taken a loan of Rs. 25000/- for 3 years, in such a way that the compound interest for 1st year, 2nd year and 3rd year is 4%, 5% and 6% per annum respectively. Find the amount to be deposited after 3 years.

6. Answer any one question:- (1X3=3)

i. Speed of A is 1 m/sec more than that of B . A reaches 2 seconds earlier than B in 180 m sprint competition. Find the speed of B in m/sec.

ii. Solve $(2x + 1) + \frac{3}{(2x+1)} = 4, \quad x \neq -\frac{1}{2}$

7. Answer any one question:- (1X3=3)

i. If $(\sqrt{a} + \sqrt{b}) \propto (\sqrt{a} - \sqrt{b})$ then show that $a + b \propto \sqrt{ab}$

ii. If $x = \sqrt{3} + \sqrt{2}$ and $y = \frac{1}{x}$, then find the value of $\left(x + \frac{1}{x}\right)^2 + \left(\frac{1}{y} - y\right)^2$.

8. Answer any one question:- (1X3=3)

i. **If $\frac{x}{y+z}=\frac{y}{z+x}=\frac{z}{x+y}$, then show that each ratio is $\frac{1}{2}$ or (-1) .**

ii. **If a, b, c are in continued proportion, then prove that $\frac{1}{b}=\frac{1}{b-a}+\frac{1}{b-c}$**

9. Answer any one question:- **(1X5=5)**

i. **Prove that in any circle, angle on the same segment are equal.**

ii. **If two tangents are drawn to a circle from a point outside it, then prove that the line segments joining the points of contacts and the external poin are equal and they subtend equal angles at the centre.**

10. Answer any one question:- **(1X3=3)**

i. **ABCD is a circumscribed quadrilateral of a circle with centre O. Show that है AB+CD=AD+BC.**

ii. **In a triangle PQR , $\angle P = 90^0$, and PS is perpendicular to QR . Then prove that**

$$\frac{1}{PS^2}-\frac{1}{PQ^2}=\frac{1}{PR^2}$$

11. Answer any one question:- **(1X5=5)**

i. **Draw a circle of radius 4 cm. Draw a tangent on this circle from an external point at a distance 9 cm from the centre of the circle.**

ii. **Draw a right angled triangle with two adjacent sides of right angle are 4 cm and 5 cm. Construct the circumcircle of this triangle.**

12. Answer any two questions:- **(3X2=6)**

i. **The difference of acute angles of a right angled triangle is** 72^0 . Determine these acute angles in circular measure.

ii. If $5sin^2\theta + 4cos^2\theta = \frac{9}{2}$, then from this relation finds the value of $tan\theta$.

iii. If $sin17^0 = \frac{x}{y}$ then show that $sec17^0 - sin73^0 = \frac{x^2}{y\sqrt{y^2-x^2}}$.

13. Answer any one question:- **(1X5=5)**

i. **The angle of elevation of the top of a pillar of height h from two points in the same horizontal line through the foot of the pillar are in the same side of the pillar are θ and φ . Find the distance between the two points.|**

ii. **Two pillars of equal height are at the points A and B on the opposite side of the road which is 120 m wide. From a point C on the line joining the foot of the pillars, the angle of elevation of the top of the pillar at A and B 60^0 and 30^0. Find AC.**

14. Answer any two questions:- **(4X2=8)**

i. **The lower part of an ice-cream is conical and upper part is hemispheric with same circular base, If the height of the cone is 9 cm and radius of the base is 2.5 cm. Find the volume of the ice-cream.**

ii. **Difference of outer and inner curved surface area of a hollow cylindrical pipe is 44 sq cm and length is 14 cm. If the volume of the material of the pipe be 99 cubic cm. Find inner and outer radius.**

iii. **If 75 buckets of water of equal measure are taken out from the water of a cubical water filled tank, then $\frac{2}{5}$ th of the water remains in the tank. If the length of one edge of the tank is 1.5 metres, then calculate quantity of water in litre that can be held in each bucket.**

15. Answer any two questions:- **(4X2=8)**

i. **Find the mode from the following data:**

Class interval	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	2	6	10	16	22	11	8	5

ii. **Find the mean by any suitable method from the following frequency distribution table:**

Class interval	85-105	105-125	125-145	145-165	165-185	185-205
Frequency	3	12	18	10	5	2

iii. **Find the median of the data from the following distribution:**

Marks obtained	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60
No. of students	8	15	29	42	60	70

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